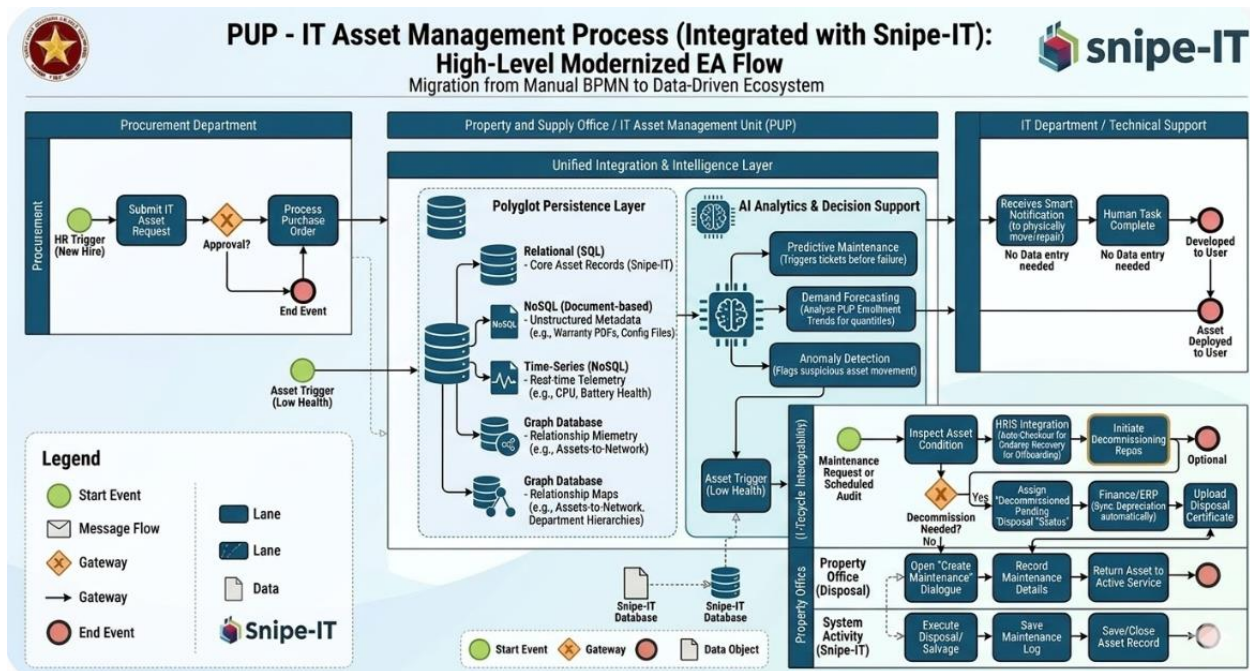


Activity 3: Data Architecture Design for IT Asset Management

Sample Solution Based on the PUP TO-BE Architecture



Introduction

This sample solution demonstrates how to complete Activity 3 deliverables using the provided TO-BE business architecture diagram for PUP's IT Asset Management Process. The architecture features a modernized, data-driven ecosystem with polyglot persistence, AI analytics, and unified integration layers.

The solution illustrates how to translate the high-level process flows into concrete data architecture deliverables that align with the enterprise data management principles taught in Module 2 and Activity 3.

Deliverable 1: Enterprise Data Model

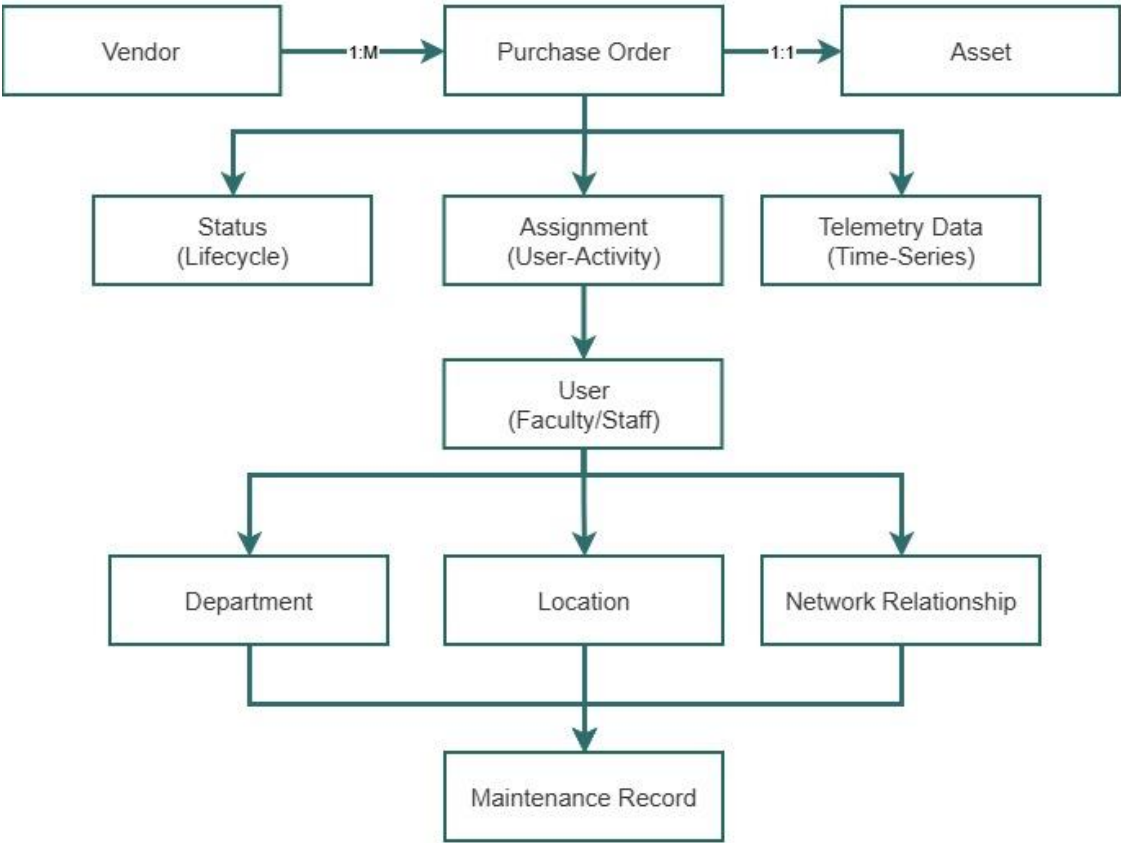
Entity Identification

Entity	Definition	Examples
Asset	Physical or digital technology item tracked by the organization	Laptop, desktop, monitor, software license, network equipment
User	Individual who uses or is assigned assets	Faculty, staff, student assistant, contractor
Department	Organizational unit that owns or uses assets	College of Engineering, Registrar's Office, HR, IT
Location	Physical or logical place where assets reside	Building, room, data center, remote site
Vendor	Company that supplies hardware or software	Dell, HP, Microsoft, Adobe
Purchase Order	Document authorizing procurement of assets	PO-2025-00123
License	Software entitlement granting usage rights	Microsoft 365, Adobe Creative Cloud, SPSS
Maintenance Record	History of repairs, service, and telemetry data	Warranty claim, repair log, CPU health readings
Assignment	Relationship between asset and user over time	Check-out record, transfer history

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Status	Current state of an asset in its lifecycle	Deployed, Maintenance, Retired, On Order
Telemetry Data	Real-time sensor readings from assets	CPU temperature, battery health, disk usage, location
Network Relationship	Connection mapping between assets and network infrastructure	Asset-to-switch, asset-to-access point, departmental hierarchy

Entity Relationship Diagram (ERD)



Cardinality Description:

Relationship	Cardinality	Description
Vendor → Purchase Order	1:M	One vendor supplies many purchase orders
Purchase Order → Asset	1:1	One purchase order creates one asset record
Asset → Status	M:1	Many assets can have the same status
Asset → Assignment	1:M	One asset has many assignments over its lifetime
User → Assignment	1:M	One user has many asset assignments
Asset → Telemetry Data	1:M	One asset generates many telemetry readings
Asset → Maintenance Record	1:M	One asset has many maintenance records
Asset → Location	M:1	Many assets can be in one location
User → Department	M:1	Many users belong to one department
Department → Department	M:1	Department hierarchy (self-referential)
Asset → Network Relationship	1:M	One asset connects to one network point

Entity Attribute Catalog

Entity	Attribute	Data Type	Description	Business Rules
Asset	AssetID	String (UUID)	Unique identifier	System-generated, immutable
Asset	AssetTag	String	Physical tag number	Format: PUP-YYYY-#####, unique
Asset	Name	String	Friendly name	Optional, max 100 chars
Asset	Category	String	Asset classification	Laptop, Desktop, Network, Software
Asset	Model	String	Manufacturer model	From approved model list
Asset	SerialNumber	String	Manufacturer serial	Must be unique, validated
Asset	PurchaseDate	Date	Date acquired	Cannot be future date
Asset	PurchaseCost	Decimal	Cost in PHP	>0, tracked for depreciation
Asset	WarrantyMonths	Integer	Warranty duration	0-60, 0 = expired
Asset	Status	Reference	Current status	From status_labels table

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Asset	AssignedTo	Reference	Current user	Null if unassigned
Asset	Location	Reference	Physical location	From locations table
Asset	Department	Reference	Owning department	From departments table
Telemetry	ReadingID	String (UUID)	Unique identifier	System-generated
Telemetry	AssetID	Reference	Associated asset	Foreign key to assets
Telemetry	MetricType	String	Type of reading	CPU_TEMP, BATTERY_HEALTH, DISK_USAGE
Telemetry	MetricValue	Decimal	Reading value	Depends on metric type
Telemetry	ReadingTimestamp	DateTime	When reading was taken	UTC timestamp
Network	RelationshipID	String (UUID)	Unique identifier	System-generated
Network	AssetID	Reference	Associated asset	Foreign key to assets
Network	NetworkNode	String	Network point identifier	Switch port, AP name

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Network	ConnectionType	String	Type of connection	WIRED, WIRELESS
Network	LastSeen	DateTime	Last connection time	Updated on each scan

Business Rules Documentation

Rule ID	Entity	Rule Description	Enforcement
BR-001	Asset	An asset cannot be assigned to an inactive user	System-enforced via HRIS integration
BR-002	Asset	Purchase cost must be > 0 PHP	System-enforced during entry
BR-003	License	License seats cannot exceed purchased quantity	System-enforced, alerts when near limit
BR-004	Assignment	An asset cannot be assigned to two active users simultaneously	System-enforced
BR-005	Asset	Retired assets must have disposal date and certification recorded	Process-enforced with workflow
BR-006	Telemetry	Health alerts trigger when metrics exceed thresholds	System-enforced, triggers predictive maintenance
BR-007	Network	Assets detected on unauthorized network segments trigger alert	System-enforced via anomaly detection

BR-008	Status	Status transitions must follow defined workflow path	System-enforced with status validation
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Deliverable 2: System Data Model Mapping

Snipe-IT Core Tables Mapping

Enterprise Entity	Snipe-IT Table	Match?	Notes
Asset	assets	✓ Full	Core asset table
User	users	✓ Full	User accounts
Department	departments	✓ Full	Department/location mapping
Location	locations	✓ Full	Physical locations
Vendor	vendors	✓ Full	Supplier/vendor records
License	licenses	✓ Full	Software licenses
Status	status_labels	✓ Full	Customizable statuses
Assignment	-	X Missing	No direct assignment history table
Maintenance Record	-	X Missing	Warranty and repair logs need custom solution

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Telemetry Data	-	X Missing	Requires time-series database integration
Network Relationship	-	X Missing	Requires graph database integration

Field-Level Mapping

Enterprise Entity	Enterprise Attribute	Snipe-IT Table	Snipe-IT Field	Match?	Notes
Asset	AssetID	assets	id	✓	Auto-increment integer
Asset	AssetTag	assets	asset_tag	✓	Customizable format
Asset	Name	assets	name	✓	Optional
Asset	Category	assets	category_id	✓	References categories table
Asset	Model	assets	model_id	✓	References models table
Asset	SerialNumber	assets	serial	✓	
Asset	PurchaseDate	assets	purchase_date	✓	

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Asset	PurchaseCost	assets	purchase_cost	✓	
Asset	WarrantyMonths	assets	warranty_months	✓	
Asset	Status	assets	status_id	✓	References status_labels
Asset	AssignedTo	assets	assigned_to	✓	References users
Asset	Location	assets	location_id	✓	References locations
Asset	Department	assets	company_id	✓	References companies
Telemetry	ReadingID	-	-	X	Not in Snipe-IT, in time-series DB
Telemetry	MetricType	-	-	X	In time-series DB
Telemetry	MetricValue	-	-	X	In time-series DB
Telemetry	ReadingTimestamp	-	-	X	In time-series DB
Network	RelationshipID	-	-	X	Not in Snipe-IT, in graph DB

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Network	NetworkNode	-	-	X	In graph DB
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Gap Analysis: Missing Enterprise Concepts

Missing Concept	Enterprise Entity	Required For	Proposed Solution
Assignment History	Assignment	Audit trail, lifecycle tracking, decommissioning verification	Custom plugin or API logging; integrate with polyglot persistence layer
Maintenance Record	Maintenance	Repair tracking, warranty claims, predictive maintenance triggers	Custom field group + integration with time-series database for telemetry
Asset Relationships	Asset	Dependencies (e.g., laptop + dock), network topology	Graph database (Neo4j) integration for relationship mapping
Telemetry Data	Telemetry	Real-time health monitoring, predictive maintenance, anomaly detection	Time-series database (InfluxDB/TimescaleDB) with API integration
Network Topology	Network	Asset-to-network mapping, location tracking, security monitoring	Graph database (Neo4j) for relationship maps
Approval Workflow	Request	Manager approval tracking for asset requests	Custom statuses + workflow engine integration

Custom Field Proposals

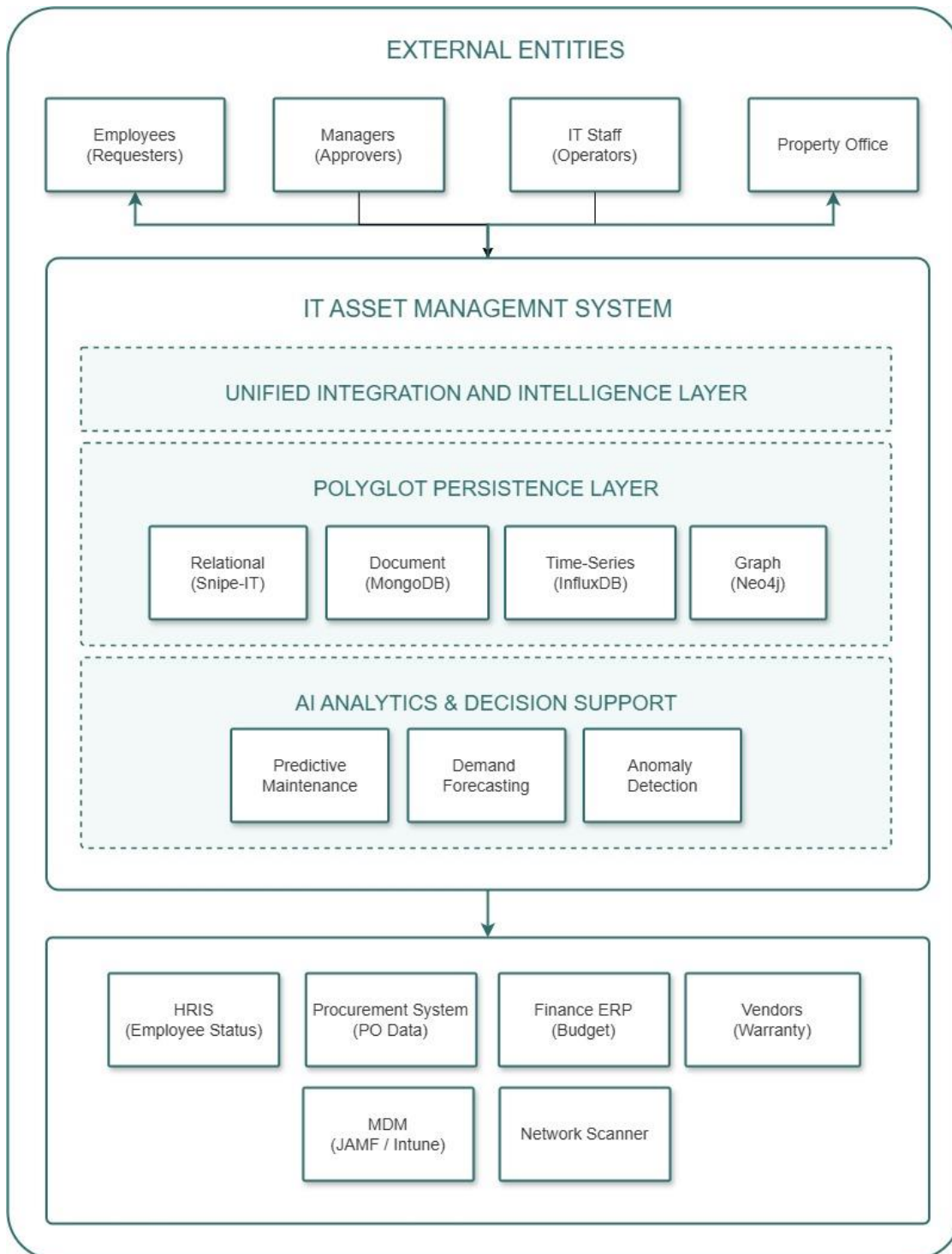
Custom Field	Applies To	Data Type	Purpose	Business Rule
Asset Condition	Asset	Dropdown	Track physical condition	Values: New, Good, Fair, Poor, For Repair
Last Audit Date	Asset	Date	Track physical inventory cycle	Update annually
PO Number	Asset	String	Link to purchase order	Format: PO-YYYY-#####
Cost Center	Asset	String	Financial allocation	From finance system
Assigned Date	Assignment	Date	Track assignment start	Auto-populated on checkout
Expected Return	Assignment	Date	For loaner devices	Optional
Health Score	Asset	Integer	Aggregate telemetry health	0-100, auto-calculated
Warranty Expiry Alert	Asset	Boolean	Flag for expiring warranty	Auto-set 30 days before expiry

Custom Status Label Proposals

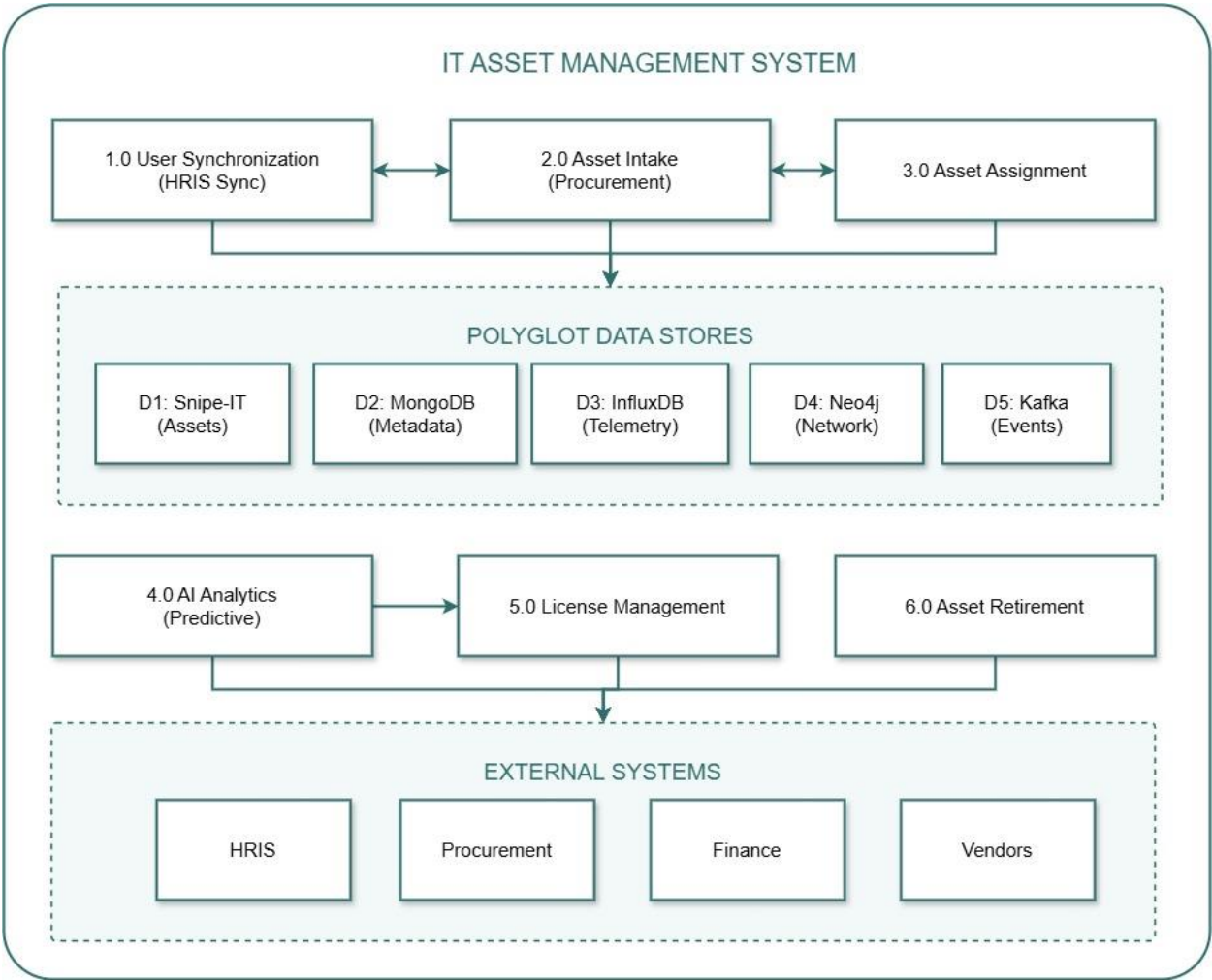
Status Label	Category	Description	Workflow Position
On Order	Asset	Ordered but not received	Before Ready to Deploy
Pending Imaging	Asset	Received, awaiting configuration	Between Received and Ready
Imaging Complete	Asset	Configured, ready for QA	Before Ready to Deploy
Awaiting Pickup	Asset	Assigned but not physically handed over	Between Assigned and Deployed
Maintenance - In Shop	Asset	Being repaired internally	During Maintenance
Maintenance - Vendor	Asset	Sent to vendor for repair	During Maintenance
Pending Return	Asset	Employee offboarded, awaiting return	Before Retired
Pending Disposal	Asset	Approved for disposal, awaiting processing	Before Disposed
Health Alert	Asset	Telemetry indicates potential failure	Triggers preventive workflow

Deliverable 3: Data Flow Diagrams (DFDs)

Level 0: Context Diagram



Level 1: Decomposed DFD



Data Store Descriptions

Store ID	Name	Technology	Purpose
D1	Asset Store	PostgreSQL (Snipe-IT)	Core asset records, assignments, user data
D2	Metadata Store	MongoDB	Unstructured metadata, warranty PDFs, config files

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Store ID	Name	Technology	Purpose
D3	Telemetry Store	InfluxDB/TimescaleDB	Real-time asset health data, CPU, battery, location
D4	Relationship Store	Neo4j	Network topology, department hierarchies, asset relationships
D5	Event Store	Kafka	Change data capture events, audit logs

Data Flow Descriptions

Flow Name	Source	Destination	Data Content	Frequency	Volume
Employee Status Update	HRIS	1.0 User Sync	Employee ID, Name, Dept, Status	Real-time (webhook)	50/day
PO Data	Procurement System	2.0 Asset Intake	PO#, Vendor, Items, Costs, ETA	Daily batch	20/day
Assignment Record	3.0 Assignment	D1, D4	Asset ID, User ID, Date, Location	Per transaction	50/day
Telemetry Stream	MDM/Network	3.0 Telemetry Ingestion	Asset ID, CPU, Battery, Health, Location	Continuous (5-min intervals)	10,000/day

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Predictive Alert	4.0 AI Analytics	IT Staff	Asset ID, Failure Probability, Recommendation	On detection	10/day
Network Topology	Network Scanner	4.0 Relationship Mapping	Asset ID, Switch Port, AP, VLAN	Daily	1,000/day
Depreciation Data	D1	Finance ERP	Asset ID, Purchase Cost, Age, Status	Monthly	500/month
Warranty Claim	6.0 Retirement	Vendors	Asset ID, Serial, Warranty Status	As needed	50/month

Deliverable 4: Data Domain Ownership Matrix

Domain Ownership Table

Domain	Description	Data Owner	Data Steward	Data Custodian	Primary System
Asset	Physical and digital assets	IT Asset Manager	Asset Team Lead	IT Operations	Snipe-IT
User	Employees and contractors	HR Director	HRIS Administrator	IT Operations	HRIS / AD
License	Software entitlements	IT Asset Manager	Software Asset Specialist	IT Operations	Snipe-IT

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Vendor	Hardware/software suppliers	Procurement Manager	Procurement Specialist	Finance Systems	Procurement System
Financial	Cost and depreciation	Finance Controller	Financial Analyst	Finance Systems	ERP
Location	Physical locations	Facilities Manager	Operations Coordinator	IT Operations	Snipe-IT + Graph DB
Telemetry	Real-time asset health	IT Operations Manager	Systems Administrator	IT Operations	InfluxDB
Network	Asset-to-network relationships	Network Manager	Network Engineer	IT Operations	Neo4j
Procurement	Purchase transactions	Procurement Manager	Procurement Specialist	Finance Systems	Procurement System
Maintenance	Repair history	IT Support Manager	Help Desk Lead	IT Operations	Snipe-IT + Custom

Data Owner Responsibilities

Role	Responsibilities
Data Owner (Business Leader)	• Define data quality standards • Approve data access requests • Resolve escalated data issues • Ensure regulatory compliance • Define retention and disposal policies • Budget for data management initiatives

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Data Steward (Operational)	• Monitor data quality daily • Execute data cleanup activities • Train users on proper data entry • Report issues to data owner • Maintain data documentation and dictionaries
Data Custodian (Technical)	• Implement technical controls • Manage database access permissions • Ensure backup and recovery procedures • Monitor system performance • Apply security patches and updates

RACI Matrix for Data Decisions

Activity	Asset Owner	Asset Steward	HR Director	IT Ops	Finance	Procurement	Network Admin
Define asset data standards	A	R	C	C	I	I	I
Enter new asset data	I	R	-	C	-	C	-
Update asset status	I	R	-	C	-	-	-
Approve asset disposal	A	C	-	I	C	-	-

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Define telemetry thresholds	C	I	-	A	-	-	R
Map network relationships	-	-	-	C	-	-	A
Resolve data quality issue	A	R	I	C	I	I	I
Grant data access	A	C	-	R	-	-	-
Define retention policy	A	C	C	I	R	I	I

(R = Responsible, A = Accountable, C = Consulted, I = Informed)

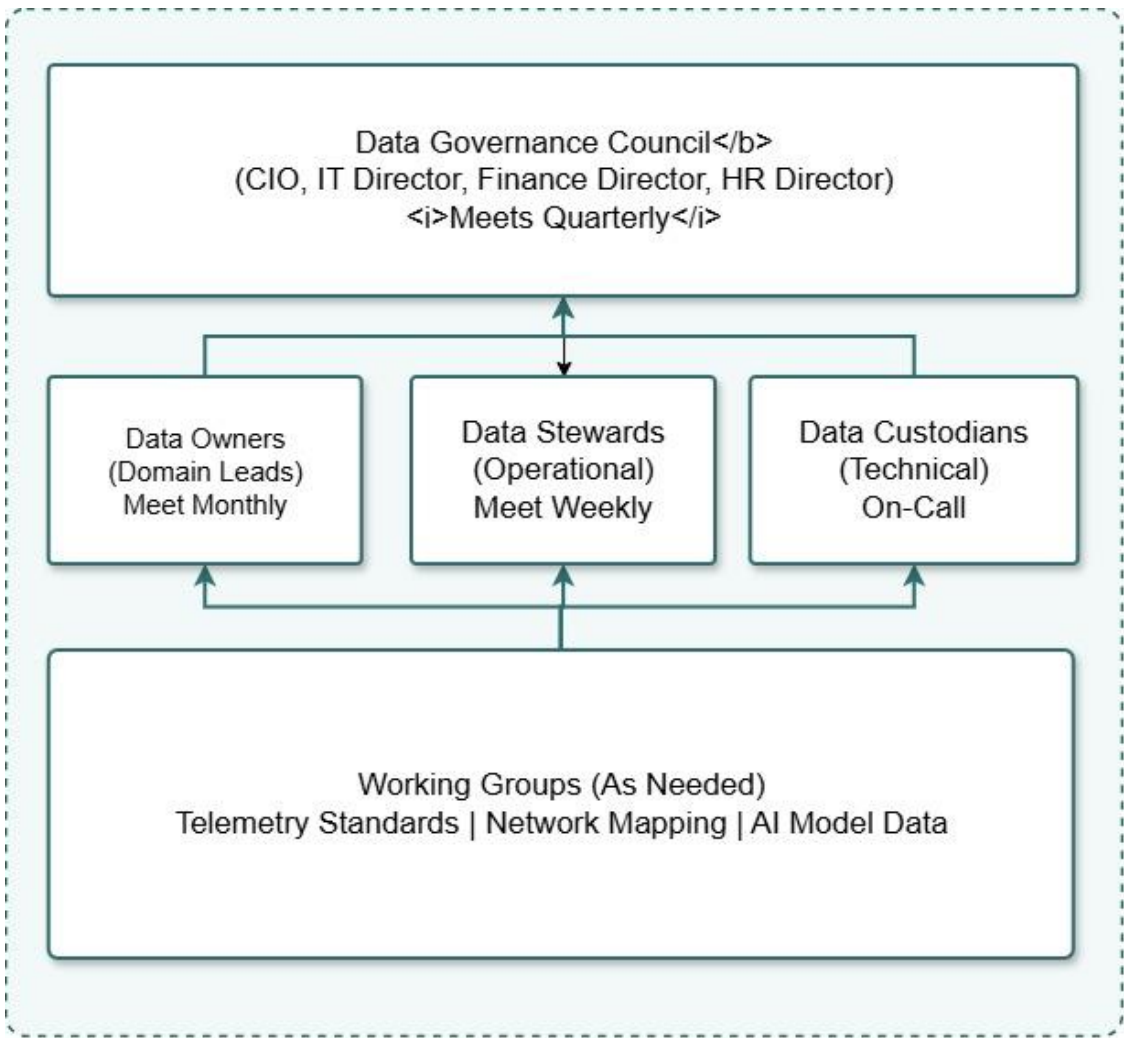
Data Issue Escalation Path

Issue Severity	Description	Escalation Path	Response Time
Critical	Data integrity breach, audit finding, security incident	Data Steward → Data Owner → IT Director	4 hours
Major	Significant quality issue affecting operations or reporting	Data Steward → Data Owner	24 hours

Issue Severity	Description	Escalation Path	Response Time
Minor	Isolated data error, formatting issue	Data Steward resolves	5 business days
Telemetry Anomaly	AI-detected unusual pattern	System Alert → Network Admin/IT Ops	Immediate

Deliverable 5: Data Governance Framework

Governance Structure



Data Policies

Policy	Description	Key Provisions
Data Quality Policy	Standards for data accuracy and completeness	• Asset records must be 99% complete • Monthly quality audits by stewards • Automated validation rules on entry • Telemetry accuracy validated against physical inventory
Data Access Policy	Who can access what data	• Role-based access control in Snipe-IT • Least privilege principle enforced • Access review every 6 months • Graph database access restricted to network team
Data Retention Policy	How long data is kept	• Active assets: indefinitely • Disposed assets: 7 years • Telemetry data: 90 days (raw), 3 years (aggregated) • Audit logs: 7 years • Network relationship data: 1 year
Data Security Policy	Protection of sensitive data	• Encryption at rest and in transit for all databases • PII data restricted to authorized users • Breach notification within 24 hours • Telemetry data anonymized for analytics
Data Classification Policy	Labeling data sensitivity	• Public: Asset tags, locations • Internal: Purchase costs, assignments • Confidential: User PII, security logs • Restricted: Financial data, audit findings

Data Standards

Standard	Description	Rule
Asset Naming Convention	How assets are named	{Type}-{Dept}-{Serial-Last-4} e.g., LAPTOP-ENG-1234
Asset Tag Format	Format for physical tags	PUP-{YYYY}-{5-digit-sequence} e.g., PUP-2025-01234
Status Values	Allowed statuses	From approved list (customized with AI-triggered health alerts)
Date Format	Standard date representation	ISO 8601: YYYY-MM-DDTHH:MM:SSZ
Currency	Standard monetary unit	PHP (Philippine Peso) with 2 decimal places
Telemetry Metric Types	Standard metric naming	ASSET.CPU.TEMP, ASSET.BATTERY.HEALTH, ASSET.LOCATION
Network Node Identifier	Format for network points	{Building}-{Floor}-{Switch}-{Port} e.g., MAIN-2-SW01-P08

Data Quality Metrics

Metric	Definition	Target	Measurement Method
Completeness	% of required fields populated	>98%	Monthly automated audit

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Accuracy	% of records matching physical inventory	>99%	Quarterly cycle count
Timeliness	% of updates within 24 hours of event	>95%	System logs, CDC monitoring
Consistency	% of records following standards	>98%	Automated validation
Uniqueness	% of records without duplicates	100%	Duplicate detection weekly
Telemetry Coverage	% of assets reporting health data	>95%	InfluxDB query
Relationship Accuracy	% of network mappings correct	>99%	Weekly network scan validation

Data Issue Management Process

Step	Description	Responsible	Timeline
1. Identification	Issue detected via validation, user report, audit	Automated/User	Continuous
2. Logging	Issue recorded in tracking system with severity	Data Steward	Within 1 hour
3. Triage	Steward assesses, assigns root cause	Data Steward	Within 4 hours
4. Resolution	Root cause fixed, data corrected	Data Steward/Custodian	Based on severity
5. Verification	Fix confirmed, validation passes	Data Steward	Within 24 hours
6. Closure	Issue documented, lessons learned	Data Steward	At resolution
7. Prevention	Process improved to prevent recurrence	Data Owner	Monthly review

Deliverable 6: Data Quality Assessment

Data Quality Baseline

Data Element	Entity	Current Quality	Issues Identified	Impact
Asset Tag	Asset	92%	Format variations, duplicates (pre-2023)	Search failures, duplicate procurement
Assignment	Asset	88%	Assets assigned to inactive users, missing return dates	Inaccurate inventory, security risk
Location	Asset	75%	Outdated, missing, inconsistent naming	Can't locate assets, audit failures
Purchase Cost	Asset	65%	Missing for older assets, estimated values	Finance doesn't trust, depreciation errors
Status	Asset	78%	Wrong statuses used, inconsistent transitions	Process confusion, AI model confusion
Department	User	85%	Some missing, transfers not updated	Reporting gaps, wrong cost centers
License Allocation	License	70%	Overallocation, underutilization, manual tracking	Compliance risk, audit findings

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Data Element	Entity	Current Quality	Issues Identified	Impact
Telemetry Data	Asset	60%	Coverage gaps for older devices, inconsistent metrics	Predictive maintenance unreliable
Network Mapping	Asset	55%	Manual updates, stale data, missing connections	Security monitoring gaps

Root Cause Analysis for Data Issues

Issue	Root Cause	Impacted Processes	Severity
Assets assigned to inactive users	No HRIS integration; manual offboarding fails	Retirement, Audit, Finance, Security	Critical
Outdated locations	No location update process; no self-service	Movement, Physical Inventory, Network	High
Missing purchase costs	Procurement not integrated; paper records	Finance, Depreciation, Budget	High
Wrong statuses	Unclear definitions; no workflow enforcement	All processes, AI analytics	Medium
Telemetry gaps	Inconsistent agent deployment; older devices	Predictive Maintenance, AI	Medium

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Issue	Root Cause	Impacted Processes	Severity
Network mapping stale	Manual updates; no automated discovery	Security, Asset Tracking	High
License overallocation	Manual tracking; no usage monitoring	Compliance, Audit	Critical

Data Improvement Targets

Data Element	Current	Target	Improvement Needed	Method
Asset Tag	92%	100%	Eliminate duplicates, enforce format	Validation rules, data cleanup, training
Assignment	88%	99%	Automate offboarding	HRIS integration, workflow automation
Location	75%	98%	Self-service updates, periodic audits	Mobile app, beacon integration, process
Purchase Cost	65%	100%	Procurement integration, historical cleanup	API integration, data entry project
Status	78%	98%	Clarify definitions, simplify options	Documentation, workflow enforcement
Telemetry Coverage	60%	95%	Deploy agents to all devices	MDM integration, agent rollout

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Data Element	Current	Target	Improvement Needed	Method
Network Mapping	55%	99%	Automated discovery	Network scanner integration, Neo4j
License Allocation	70%	100%	Usage monitoring, automated true-up	Integration with usage tools

Data Cleanup Plan

Activity	Responsible	Timeline	Resources Needed
Correct asset tag duplicates	Asset Steward	Month 1	30 hours
Update locations for top 1,000 assets	Help Desk + Asset Team	Month 1-2	80 hours
Research missing purchase costs	Procurement + Finance	Month 2	60 hours
Reconcile license allocations	Software Asset Specialist	Month 2-3	40 hours
Deploy telemetry agents	IT Operations	Month 2-4	2 FTE, MDM integration
Implement network auto-discovery	Network Team	Month 3-4	1 FTE, scanner tools

Activity	Responsible	Timeline	Resources Needed
Full physical inventory audit	Asset Team + Help Desk	Month 3	3 weeks, all hands
Historical data cleanup	Data Stewards	Month 4	Ongoing

Deliverable 7: Traceability Matrix

Pain Point to Data Solution Traceability

Pain Point ID	Pain Point Description	Data Solution	Deliverable Reference
P1	Procurement-to-inventory delay (3-5 days)	Procurement API integration; automated asset creation upon PO approval	DFD 2.0, System Mapping, Custom Field: PO Number
P3	Poor mobile experience for scanning	Mobile-optimized data entry; offline capability; beacon integration	DFD 3.0, Custom Fields, Telemetry integration
P5	Software license over-purchasing	License usage tracking; allocation automation; compliance dashboard	Domain Ownership, Governance, DFD 5.0
P7	Asset non-return after offboarding	HRIS integration; automated status updates; return workflow	DFD 1.0, System Mapping, Custom Status: Pending Return

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Pain Point ID	Pain Point Description	Data Solution	Deliverable Reference
P9	Finance maintains separate spreadsheets	Depreciation data export; API integration with ERP; real-time cost visibility	DFD, Data Standards, Telemetry for asset value
P12	Audit findings on inventory accuracy	Complete audit trail; assignment history tracking; network mapping validation	Custom Field, Governance, Graph DB for relationships
P15	Predictive maintenance failures	Telemetry ingestion; AI anomaly detection; health score tracking	DFD 4.0, Time-series DB, Governance

TO-BE Process to Data Support Traceability

TO-BE Process	Data Requirements	Data Solution	Deliverable Reference
1.0 Procurement Intake	PO data, vendor data, cost data, ETA	Procurement API integration; custom PO field; automated asset creation	System Mapping, DFD 2.0, Custom Fields
2.0 Asset Assignment	User data, asset data, assignment tracking, location	HRIS sync; role-based templates; assignment history; mobile check-out	Enterprise Model, DFD 3.0, Graph DB for location

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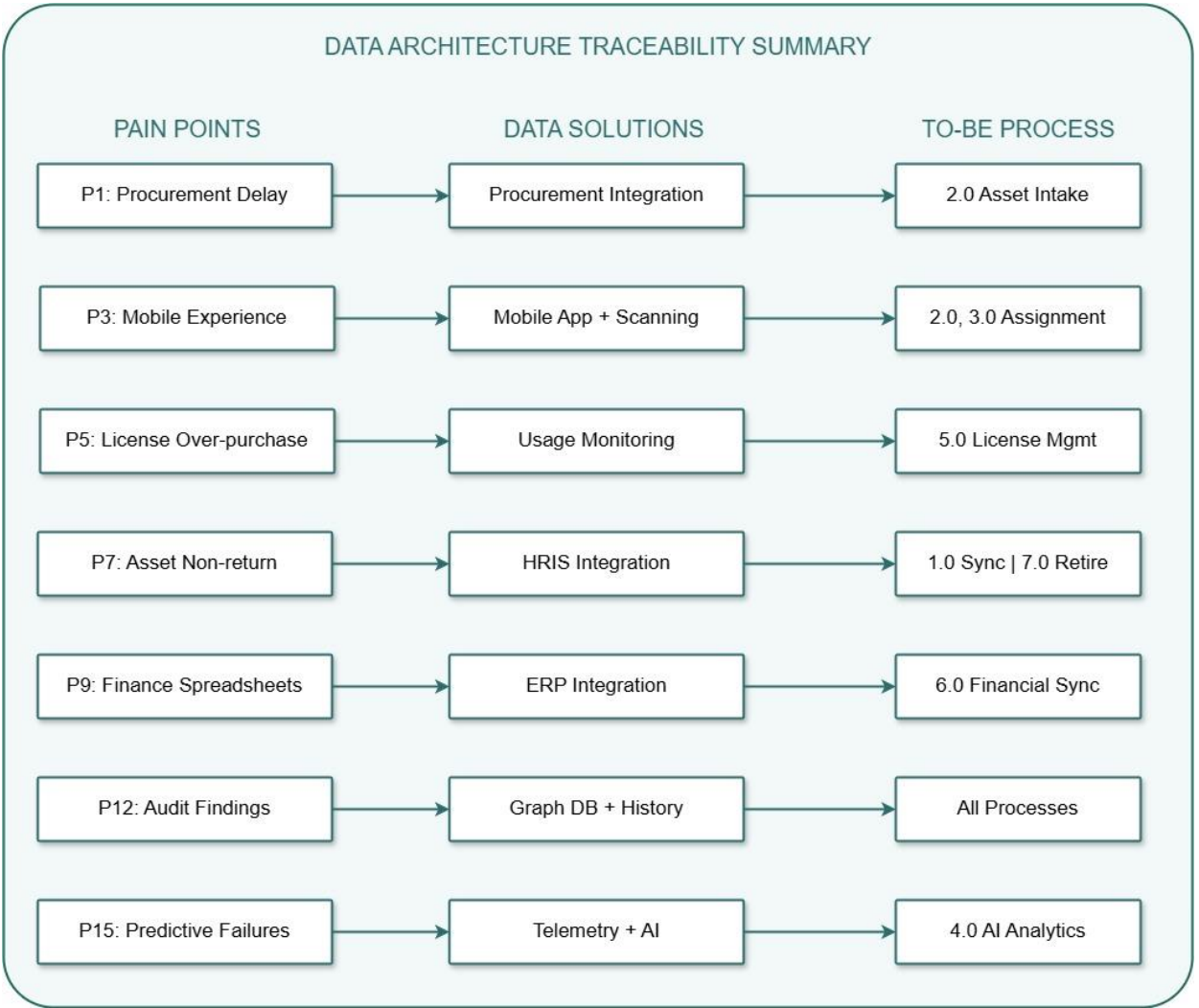
TO-BE Process	Data Requirements	Data Solution	Deliverable Reference
3.0 Telemetry Ingestion	Health metrics, location data, usage patterns	MDM integration; agent deployment; time-series database	DFD 3.0, InfluxDB, Telemetry Standards
4.0 AI Analytics	Historical data, real-time telemetry, maintenance records	Machine learning models; anomaly detection; predictive alerts	DFD 4.0, AI Integration, Governance
5.0 Network Mapping	Asset connections, switch ports, AP associations	Automated network scanner; graph database; relationship mapping	DFD 4.0, Neo4j, Network Standards
6.0 License Management	License allocation, usage data, renewal dates	Usage monitoring integration; compliance reporting; automated alerts	Domain Ownership, DFD 5.0, Governance
7.0 Asset Retirement	Disposal data, sanitization certification, financial close	Retirement workflow; document storage; ERP integration	Custom Fields, Governance, DFD 6.0

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Data Solution Justification Summary

Data Solution	Cost/Effort	Pain Points Addressed	Processes Enabled	Priority
HRIS Integration	Medium	P7, P12	User Sync, Retirement, Audit	Critical
Procurement Integration	Medium	P1, P9	Intake, Finance, Budget	Critical
Telemetry & Time-Series DB	High	P15, P12	Predictive Maintenance, AI	Important
Graph DB (Neo4j)	Medium	P12, Network Security	Network Mapping, Audit	Important
Mobile App with Scanning	High	P3, P12	Assignment, Movement, Audit	Important
License Usage Integration	Medium	P5	License Management, Compliance	Important
AI Analytics Platform	High	P15, P12, P5	All processes	Strategic
Assignment History Tracking	Low	P12	All processes	Tactical

Traceability Summary Matrix



Conclusion

This sample solution demonstrates how the TO-BE business architecture for PUP's IT Asset Management Process translates into concrete data architecture deliverables. The key innovations reflected in this solution include:

- 1. **Polyglot Persistence:** The architecture leverages multiple database technologies— PostgreSQL (Snipe-IT) for core asset records, MongoDB for unstructured metadata, InfluxDB for telemetry time-series data, and Neo4j for network relationship graphs.

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2. **AI Analytics Integration:** Data flows from telemetry ingestion through AI analytics to predictive maintenance alerts, creating a closed-loop intelligent asset management system.
3. **Unified Integration Layer:** The architecture coordinates data flows across HRIS, procurement, finance, MDM, and network systems, ensuring consistency and timeliness.
4. **Comprehensive Governance:** Clear ownership, policies, and quality metrics ensure that data remains reliable over time.

The traceability matrix ensures that every data decision traces back to documented pain points and supports the TO-BE processes. This alignment between business needs and data architecture is the foundation for a successful transformation.